

PROBLEM STATEMENTS

Three build-ready challenges across Healthcare and Cybersecurity. Each statement defines the problem and the inputs available. The approach is yours: rule-based scoring, machine learning, or a hybrid.

HEALTHCARE

PS-01

Patient Follow-up Risk Predictor

Hospitals lose track of patients who need follow-up visits. A missed follow-up is not just an empty slot — it is a treatment left incomplete and a condition left to worsen. Build a system that predicts which patients are most likely to miss their next follow-up, and ranks them so staff can intervene beforehand. A risk score on its own is not enough: every prediction must carry the reasons that produced it, because a nurse acts on reasons, not on a number.

INPUT

- Previous appointment history
- Age
- Number of missed appointments
- Distance from hospital
- Treatment duration
- Appointment frequency

AI OPPORTUNITY

Model or rules — your call

Train a simple ML classifier on historical attendance data, or build a transparent rule-based scoring system with weighted factors. Both are valid. What is judged is whether the ranking is defensible and whether the explanation genuinely follows from the inputs.

CYBERSECURITY

PS-02

Phishing Attack Investigation Platform

An employee receives a mail from **security@paypa1-login.com**, subject “**Your account will be suspended!**”, pointing to **http://paypa1-login.com/verify** — and has seconds to decide whether to trust it. Build a system that analyses the message and answers that question with evidence rather than a bare verdict, scoring the risk and listing every indicator that fired.

ANALYSE

- Sender address
- Domain
- URL
- Urgency language
- Suspicious keywords
- Domain similarity to known brands
- Links
- Attachments

BONUS

Automatic incident report

Generate a shareable incident report from the analysis — verdict, evidence, indicators of compromise, and a recommended action for the security team, produced without any manual write-up.

Cyber Incident Prioritization Engine

A security team receives 100 alerts in a single shift — failed logins, malware detections, port scans, data exfiltration, suspicious emails, brute-force attempts. They cannot investigate everything simultaneously, and the loudest alert is rarely the most dangerous one. Build a system that decides which incident the team should investigate first.

SCORING FACTORS

- Severity
- Asset importance
- Number of affected users
- Data sensitivity
- Attack confidence
- Business impact

CHALLENGE

Build the scoring engine

The queue is the product. Feed the engine a mixed batch of alerts and return a ranked list, not a single verdict — then justify why the incident at the top outranks the one below it. This tests algorithmic thinking: weight design, normalisation across factors on different scales, and sensible tie-breaking.